

Polyhedral cones = finitely generated cones

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In this note we show that polyhedral cones and finitely generated cones are the same objects. Specifically we show:

Theorem 1 (*Finitely generated cones and polyhedral cones are equivalent*)
Given a matrix $A \in \mathbb{R}^{m \times n}$, there exists a finite set $E \subset \mathbb{R}^n$ such that:

$$\{x \in \mathbb{R}^n : Ax \leq 0_m\} = \text{cone}(E), \quad (1)$$

and conversely, given a finite set $E \subset \mathbb{R}^n$, there exists a matrix $A \in \mathbb{R}^{m \times n}$ such that (1) is satisfied. ■

One direction of Theorem 1 is a corollary of the projection theorem for polyhedral cones, and this has been shown in note 1.(D):

Corollary 1 (*A finitely generated cone is a polyhedral cone*)
Finitely generated cones are polyhedral cones. ■

Here we show the other direction, *i.e.*, we show polyhedral cones are finitely generated. Therefore let $C = \{x \in \mathbb{R}^n : Ax \leq 0_m\}$ be a polyhedral cone, where $A \in \mathbb{R}^{m \times n}$, and let $a_1, \dots, a_m \in \mathbb{R}^n$ be the rows of A . We must show that there exists a finite set $E \subset \mathbb{R}^n$ such that (1) holds.

The argument uses results on duality for convex cones (from note 1.(E)):

Fact 1 *The dual C° of the polyhedral cone C is the set:*

$$C^\circ := \{\alpha \in \mathbb{R}^n : \alpha^T x \leq 0 \text{ holds for all } x \in C\},$$

and one has $C^\circ = \text{cone}(\{a_1, \dots, a_m\})$. Furthermore, the dual of C° is again C , *i.e.*, one has $(C^\circ)^\circ = C$. ■

Fact 1 gives $C^\circ = \text{cone}(\{a_1, \dots, a_m\})$. Hence the dual of C is finitely generated. Applying Corollary 1 to C° then shows C° is a polyhedral cone, *i.e.*, there exists a matrix $D \in \mathbb{R}^{p \times n}$ such that $C^\circ = \{\alpha \in \mathbb{R}^n : D\alpha \leq 0_p\}$.

Let $d_1, \dots, d_p$ be the rows of D . Applying Fact 1 again to the polyhedral cone C° gives $(C^\circ)^\circ = \text{cone}(\{d_1, \dots, d_p\})$. Combined with $C = (C^\circ)^\circ$ this shows C is finitely generated by the rows of D . This completes the proof of Theorem 1.